Status	Finished
Started	Thursday, 15 October 2026, 8:24 PM
Completed	Thursday, 15 October 2026, 8:32 PM
Duration	8 mins 6 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Quick Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It allows for faster searching
b. It increases the number of comparisons needed
c. It makes the search operation slower
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **3**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It sorts data using a single pass
b. It does not require recursion
c. It is based on the divide-and-conquer approach
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Bubble Sort
b. Linear Search
c. Binary Search
d. Merge Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting makes it harder to search for items
c. Sorting decreases the efficiency of selection operations
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **6**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It complicates the analysis process
b. It makes the analysis slower
c. It allows for quicker identification of duplicates
d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Selection sort
- c. Quick sort
- d. Heap sort

The correct answer is:
Merge sort

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Merging
- c. Sorting
- d. Complexity

The correct answer is:
Complexity

Question **9**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Bubble Sort
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **10**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Heap sort

b. Quick sort

c. Bubble sort

d. Insertion sort

The correct answer is:
Insertion sort

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion

b. It only works on small datasets

c. It simplifies the sorting process by using only one pass

d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **12**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Greedy method

b. Dynamic programming

c. Divide and conquer

d. Backtracking

The correct answer is:
Divide and conquer

Question **13**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **14**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It divides the list into sublists
- b. It bubbles up the largest element in each pass
- c. It uses the heap data structure
- d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **15**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Searching
- b. Merging
- c. Rearranging
- d. Sorting

The correct answer is: Sorting